WHO PAYS FOR THE EXPERIENCE? A REVIEW OF HOW DYNAMIC PRESCRIPTIVE ANALYTICS SYSTEMS LEARN

Online Appendix: Concept Matrix of the 173 Papers in the Review Corpus

A cross marks a characteristic of the concept matrix (Table 1 of the paper) that a paper has. Source of experience, exploration constraints and named cost bearers can take several values. The model–reality gap is coded for papers whose systems learn from recorded data or in a model. Cost reporting, quantified cost, authority over exploration and named cost bearers refer to the latest source of experience a system is designed to learn from, or to an earlier exploring trial or production phase if only passive updating follows it; quantified cost applies only where a cost is quantified. A dimension without a cross has no value for that paper (for instance, no simulator, no constraint, or an unquantified cost). The last row gives the number of papers with each characteristic.

| | | **Input** | | | | | **Throughput** | | | | |

| | | **Input** | | | | | **Throughput** | | | | |

		Input					Throughput										Output						Adaptation mechanisms									Economic accounting						Social accounting				
		Source of experience					Techniques				Simulator role			Exploration strategy					Exploration constraints				Decision output			Execution		Learning after deployment			Model–reality gap				Cost reporting			Quantified cost				Authority over exploration			Named cost bearers		
#	Paper	Recorded randomized experiments	Recorded past decisions	Model of the environment	Trial before rollout	Production	Deep RL	Other RL	Bandits	Bayesian or optimal learning	Training environment	Stand-in for evaluation	Both	Random perturbation	Optimism	Posterior sampling	Information value	Fixed design or sequential test	No exploration	Budget	Safety	Operational limits	Fairness or ethics	Single quantity	Choice from a set	Combined plan	By the system	By a human decision-maker	Fixed policy	Passively adaptive	Actively adaptive	Not discussed	Acknowledged or assumed away	Validated on real data	Corrected on-line	Not discussed	Mentioned	Quantified	Immediate cost of exploration	Net value of exploration	Cost of learning	Loss from model error	Algorithm	Operator	Nobody	Operator	Real units	Third parties
41	Eilers et al. (2014)					×		×				×		×											×		×								×							×				
42	Fang et al. (2024)		×			×		×								×								×			×						×			×							×			
43	Feit and Berman (2019)	×			×			×				×						×						×			×					×				×							×		×	
44	Ferreira et al. (2018)					×		×				×				×								×			×									×							×			
45	Ferreira and Mower (2023)				×	×				×		×					×					×			×		×									×				×			×		×	
46	Gao et al. (2023)		×	×		×	×						×	×								×			×		×			×					×								×			
47	Geevers et al. (2024)			×			×				×			×								×			×		×			×					×								×			
48	Ghavamipoor and Hashemi Golpayegani (2020)					×		×				×		×										×			×									×								×			
49	Gijsbrechts et al. (2022)			×			×				×			×										×			×			×					×								×			
50	Gottschlich (2013)					×		×				×						×						×			×									×								×		×	
51	Grant et al. (2020)					×		×				×			×									×			×									×								×			
52	Groeneveld et al. (2024)		×	×		×	×						×	×										×			×		×							×								×			
53	M. Guo et al. (2026)					×		×				×			×									×			×									×								×			
54	Gönsch and Hassler (2016)			×			×				×							×						×			×			×					×								×			
55	van der Haar et al. (2026)			×			×				×			×								×			×		×			×					×								×			
56	Harsha et al. (2025)			×			×				×			×										×			×			×					×								×			
57	Hausenblas et al. (2025)		×				×					×							×			×			×		×			×					×								×			
58	Hauser et al. (2014)	×		×		×		×				×					×					×			×		×			×					×								×			
59	He et al. (2026)					×		×							×									×			×			×						×								×			
60	Heinold et al. (2023)			×			×				×				×									×			×			×					×								×			
61	van Hezewijk et al. (2023)			×			×				×			×								×			×		×			×		×			×								×			
62	Hu et al. (2024)		×			×		×								×								×			×			×					×								×			
63	Iyengar and Singal (2024)					×		×				×				×								×			×			×					×								×			
64	Jackson (2022)			×			×				×			×										×			×			×						×								×			
65	Janus et al. (2024)					×		×				×			×									×			×								×								×			
66	Jia et al. (2026)					×		×				×							×		×				×		×									×								×			
67	C. Jiang et al. (2023)			×			×				×			×							×				×		×									×								×			

		Input					Throughput										Output				Adaptation mechanisms				Economic accounting				Social accounting			
		Source of experience					Techniques				Simulator role			Exploration strategy					Exploration constraints				Decision output			Execution	Learning after deployment			Model–reality gap				Cost reporting			Quantified cost				Authority over exploration			Named cost bearers		
#	Paper	Recorded randomized experiments	Recorded past decisions	Model of the environment	Trial before rollout	Production	Deep RL	Other RL	Bandits	Bayesian or optimal learning	Training environment	Stand-in for evaluation	Both	Random perturbation	Optimism	Posterior sampling	Information value	Fixed design or sequential test	No exploration	Budget	Safety	Operational limits	Fairness or ethics	Single quantity	Choice from a set	Combined plan	By the system	By a human decision-maker	Fixed policy	Passively adaptive	Actively adaptive	Not discussed	Acknowledged or assumed away	Validated on real data	Corrected on-line	Not discussed	Mentioned	Quantified	Immediate cost of exploration	Net value of exploration	Cost of learning	Loss from model error	Algorithm	Operator	Nobody	Operator	Real units	Third parties
68	H. Jiang et al. (2024)			×			×				×			×										×		×		×			×			×					×					
69	Jin et al. (2024)			×			×				×			×							×			×		×		×			×				×					×					
70	Jin et al. (2026)			×			×				×			×										×		×		×			×			×						×					
71	Joo et al. (2019)				×				×			×				×				×					×		×			×					×			×			×				×		
72	Kafaei et al. (2024)			×			×				×			×										×		×		×			×				×					×					
73	Kasa and Rajan (2021)					×			×			×			×									×		×		×							×					×					
74	Ke et al. (2025)			×		×	×					×		×							×				×		×			×				×					×					
75	B.-G. Kim et al. (2016)					×		×				×		×							×			×		×			×					×					×					
76	D. Kim et al. (2026)			×			×				×			×										×		×		×			×				×					×					
77	Klein (2021)					×		×				×		×										×		×			×					×					×			×		×
78	Kokkodis and Ipeirotis (2021)		×	×		×		×			×			×											×		×		×					×						×				
79	Kosolwattana et al. (2026)					×			×			×			×										×		×			×					×					×					
80	Koster et al. (2022)	×		×			×				×							×		×					×		×				×				×					×				×	
81	Kotecha and del Rio Chanona (2025)			×			×				×			×											×		×			×				×					×					
82	Kouvelis et al. (2025)			×			×				×			×											×		×			×					×					×					
83	Kropp et al. (2019)			×				×			×			×											×		×			×				×					×					
84	Kullman et al. (2022)			×			×				×			×											×		×			×				×					×					
85	Kullman et al. (2026)			×			×				×			×										×		×			×				×					×					
86	Kwon and Lee (2024)			×			×				×			×										×		×			×					×					×					
87	Lagos et al. (2025)					×			×			×				×									×		×			×					×					×					
88	Lee and Lee (2021)			×			×				×			×										×			×				×				×					×					
89	X. Li et al. (2012)			×		×		×				×		×										×			×				×				×					×					
90	M. Li et al. (2019)			×			×				×			×											×		×				×				×					×					
91	T. Li et al. (2026)		×	×		×	×					×		×											×		×		×							×					×					
92	Liang et al. (2022)			×			×				×			×											×			×					×					×					
93	Liberali and Ferecatu (2022)	×		×		×			×			×				×								×		×			×					×					×					
94	Liebman et al. (2019)					×		×				×		×											×		×			×					×					×					
95	Lin et al. (2018)			×			×				×			×							×				×		×			×					×					×					

		Input					Throughput										Output						Adaptation mechanisms						Economic accounting			Social accounting					
		Source of experience					Techniques				Simulator role			Exploration strategy					Exploration constraints				Decision output			Execution			Learning after deployment			Model–reality gap				Cost reporting			Quantified cost				Authority over exploration			Named cost bearers		
#	Paper	Recorded randomized experiments	Recorded past decisions	Model of the environment	Trial before rollout	Production	Deep RL	Other RL	Bandits	Bayesian or optimal learning	Training environment	Stand-in for evaluation	Both	Random perturbation	Optimism	Posterior sampling	Information value	Fixed design or sequential test	No exploration	Budget	Safety	Operational limits	Fairness or ethics	Single quantity	Choice from a set	Combined plan	By the system	By a human decision-maker	Fixed policy	Passively adaptive	Actively adaptive	Not discussed	Acknowledged or assumed away	Validated on real data	Corrected on-line	Not discussed	Mentioned	Quantified	Immediate cost of exploration	Net value of exploration	Cost of learning	Loss from model error	Algorithm	Operator	Nobody	Operator	Real units	Third parties
96	Y. Liu et al. (2017)		×				×												×					×			×	×				×			×							×				
97	Y. Liu et al. (2020)			×			×				×			×											×			×				×			×						×					
98	J. Liu et al. (2023)		×	×		×	×				×			×								×		×			×		×				×			×							×				
99	X. Liu (2023)	×					×											×			×				×			×					×			×							×				
100	J. Liu et al. (2025)			×			×				×			×								×			×		×				×				×							×				
101	X. Liu et al. (2025)			×			×				×			×											×		×				×				×							×				
102	Lu et al. (2018)			×				×			×			×										×			×					×			×							×				
103	Mandania et al. (2026)		×			×			×			×		×											×		×			×				×							×			×	×	
104	McKee et al. (2023)	×		×			×				×			×											×		×				×				×							×				
105	Misra et al. (2019)					×			×			×			×								×			×		×									×					×			×	×	
106	Moazeni et al. (2020)				×					×		×					×			×					×		×									×					×			×		
107	De Moor et al. (2022)			×			×				×			×										×			×				×				×							×				
108	Mousa et al. (2024)			×			×				×			×							×					×		×					×			×							×				
109	Nakabi and Toivanen (2021)			×			×				×			×											×		×					×			×							×				
110	Negoescu et al. (2011)				×					×		×					×			×				×			×									×					×			×		
111	Negoescu et al. (2018)			×		×				×			×				×							×			×					×				×					×			×		
112	Oroojlooyjadid et al. (2022)			×			×				×			×										×			×					×				×							×				
113	Pahr et al. (2026)			×			×				×			×											×		×					×				×					×			×		
114	Pahr and Grunow (2026)			×			×				×			×											×		×				×					×							×				
115	P. Li and Tuzhilin (2026)		×			×	×					×							×						×		×						×			×							×				
116	Parishad et al. (2025)			×			×				×			×										×			×					×				×							×				
117	Piri et al. (2022)			×		×			×		×					×								×			×					×				×							×				
118	Qian et al. (2024)		×			×	×					×		×						×				×			×					×				×							×				
119	Qin et al. (2020)		×			×	×					×							×					×			×						×			×							×				
120	Rachman et al. (2026)			×			×				×			×											×			×					×				×							×				
121	Rafieian (2023)	×		×				×			×								×					×			×						×			×							×				
122	Schauer and Schnurr (2022)			×		×		×			×			×						×				×			×						×			×							×				
123	Schwartz et al. (2017)					×			×			×				×									×			×						×				×					×			×		

		Input					**Throughput**														**Output**		**Adaptation mechanisms**						**Economic accounting**				**Social accounting**				
		Source of experience					Techniques				Simulator role			Exploration strategy							Exploration constraints				Decision output			Execution			Learning after deployment			Model–reality gap				Cost reporting			Quantified cost				Authority over exploration			Named cost bearers		
#	**Paper**	Recorded randomized experiments	Recorded past decisions	Model of the environment	Trial before rollout	Production	Deep RL	Other RL	Bandits	Bayesian or optimal learning	Training environment	Stand-in for evaluation	Both	Random perturbation	Optimism	Posterior sampling	Information value	Fixed design or sequential test	No exploration	Budget	Safety	Operational limits	Fairness or ethics	Single quantity	Choice from a set	Combined plan	By the system	By a human decision-maker	Fixed policy	Passively adaptive	Actively adaptive	Not discussed	Acknowledged or assumed away	Validated on real data	Corrected on-line	Not discussed	Mentioned	Quantified	Immediate cost of exploration	Net value of exploration	Cost of learning	Loss from model error	Algorithm	Operator	Nobody	Operator	Real units	Third parties
124	Schütz and Kolisch (2012)			×				×			×			×										×			×	×			×			×							×					
125	Simchi-Levi et al. (2022)					×			×			×			×											×	×			×					×				×				×			×	
126	Skandari and Shechter (2021)			×	×					×			×				×							×			×			×					×				×				×				×
127	Soeffker et al. (2019)			×				×			×				×									×			×			×		×			×							×					
128	Song and Sun (2024)	×			×		×									×				×	×			×			×			×				×				×			×			×			×	
129	Song et al. (2025)		×				×							×							×			×			×			×			×		×				×				×				
130	Stamey et al. (2026)				×				×			×						×		×				×			×			×					×				×				×			×	×
131	Stein et al. (2025)			×			×				×			×						×				×			×			×		×			×							×					
132	Sun et al. (2026)			×			×				×			×										×			×			×			×			×							×					
133	Talebian et al. (2014)					×			×			×					×								×		×			×					×				×				×			×	
134	Tang et al. (2025)					×			×			×			×								×		×		×			×					×				×				×				×
135	Teck et al. (2025)			×			×				×			×										×			×			×		×			×							×					
136	Temizöz et al. (2025)			×			×				×							×		×				×			×			×		×			×							×					
137	Temizöz et al. (2026)			×		×	×					×							×					×			×			×				×				×				×				×	×
138	Tian et al. (2022)	×		×	×					×	×					×								×			×			×				×				×			×			×			×	
139	Urban et al. (2014)					×			×							×									×		×			×					×				×				×				
140	Valogianni et al. (2020)				×	×			×			×						×							×		×			×					×				×				×				
141	Verleijsdonk et al. (2024)			×			×				×			×										×			×			×		×			×							×					
142	Waisman et al. (2025)				×				×			×				×									×		×			×					×				×				×			×	
143	H. Wang et al. (2022)			×				×			×			×										×			×			×					×							×					
144	W. Wang et al. (2023)	×		×			×				×				×							×			×		×			×			×			×							×					
145	B. Wang et al. (2024)		×			×				×		×				×						×			×		×			×				×				×				×					
146	Y. Wang et al. (2024)					×			×			×			×								×		×		×			×					×				×				×				×
147	Y. Wang et al. (2026)		×	×			×				×			×										×			×			×		×			×							×					
148	C.-A. Wang et al. (2026)			×			×				×			×										×			×			×				×				×				×					
149	Z. Wang et al. (2026)					×		×				×				×									×		×			×					×				×				×			×	
150	J. Xie et al. (2023)			×		×	×				×			×										×			×			×			×		×								×				
151	Y. Xie et al. (2026)			×		×	×				×			×										×			×			×			×		×								×				

| | | **Input** | | | | | **Throughput** | | | | |

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